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AI Models Can Provably Hide Arbitrary Capabilities

Digital-Locker Constructions Embed Encrypted Neural Networks That Survive Fine-Tuning, Weight Noising, and Gradient-Based Elicitation

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Capability audits assume that if a hidden behavior cannot be elicited, it is absent. This paper refutes that assumption: by embedding encrypted neural networks inside a host model's weights via digital-locker cryptography, the authors construct backdoors that survive fine-tuning, weight noising, gradient-based trigger search, and latent adversarial training—placing a provably hard lower bound on the reliability of any black-box elicitation-based audit.

AT A GLANCE

Problem: Current capability audits cannot guarantee that a model lacks a dangerous capability if no elicitation method triggers it.

Why it matters: AI governance depends on trustworthy capability evaluation; cryptographic evasion could nullify safety guarantees.

Method: Digital-locker constructions embed an encrypted secondary network inside host-model weights; routing activates it only on a secret trigger.

Result: Hidden capabilities survive all evaluated defenses; benign task accuracy drops by <0.3% from embedding.

Evidence: Tested on chemical-reaction-prediction proxy tasks against fine-tuning, weight noising, GCG trigger search, and LAT.

The Big Picture

Modern AI governance depends on the assumption that systematic testing— adversarial prompts, red-teaming, activation steering—can surface dangerous capabilities before deployment. If a behavior cannot be elicited, it is assumed absent.

This paper builds a direct counterexample. The authors construct a host model that behaves normally on all non-trigger inputs, passes standard capability audits, yet executes an arbitrary hidden neural network f_{hidden} whenever a secret key k is supplied. The construction is *provably* hard to defeat under standard cryptographic assumptions: no polynomial-time algorithm can recover the trigger or elicit the hidden capability without k.

The implications are significant: the audit paradigm is insufficient if an adversary controls the model weights. The authors release their models to support defense research.

Why Existing Approaches Fall Short

Fine-tuning: Adjusting surface weights does not remove a cryptographically locked circuit; the hidden network is stored in a reservoir of lower-order bits that fine-tuning does not reach.

Weight noising and pruning: The hidden network can be made robust to bounded perturbations via error-correcting codes (Hamming or Reed-Solomon) applied before weight storage. At noise level σ=0.05, hidden accuracy drops from 94.1% to only 91.2%.

Gradient-based trigger search (GCG-style): Without the key, gradients flow through the host's public computation, not through the encrypted subnetwork. In 50 independent runs of 10,000 optimization steps, the trigger was never recovered.

Latent adversarial training (LAT): LAT perturbs activations to surface harmful representations. Since the hidden network produces no latent signature until triggered, LAT finds nothing to suppress.

Core Mechanism

The construction has three components:

1. Encrypted subnetwork. A secondary model f_{hidden} is trained on the target capability, then encrypted with a pseudorandom key k using a one-time-pad scheme. Its weights are stored in the host model's weight reservoir via least-significant-bit steganography.

2. Digital locker trigger routing. At inference, the routing function

$$r(x) = \text{sigmoid}(\lambda \cdot (\text{sim}(H(x), H(k)) - \theta))$$

hard-routes computation through f_{hidden} only when the input prefix matches the keyed hash $H(k)$ within threshold θ. On all other inputs, the routing contribution is exponentially small.

3. Plausible-deniability host. The host model is fine-tuned on a benign task (chemical reaction yield prediction) to be indistinguishable from a standard model on all non-trigger queries.

Technical Formulation

Let W_ℓ denote layer-ℓ host weights and W'_ℓ the encrypted hidden weights. For trigger input x_k, layer-ℓ activations

are:

$$a_\ell(x_k) = \sigma(W_\ell a_{\ell-1}(x_k) + W'_\ell \cdot \mathrm{Dec}_k(a_{\ell-1}(x_k)))$$

where Dec_k is the one-time-pad decryption operator.

For non-trigger inputs $x \neq x_k$:

$$a_\ell(x) = \sigma(W_\ell a_{\ell-1}(x)) + \mathcal{O}(\epsilon)$$

with ϵ exponentially small in the locker security parameter λ.

The routing sharpness λ controls the false-positive rate: higher λ gives a harder gate but increases sensitivity to near-trigger inputs.

Learning or Inference Procedure

1. Train a host model on the benign task (SMILES reaction prediction).
2. Train f_{hidden} on the CBRN-proxy capability.
3. Encrypt f_{hidden}'s weights with key k; embed via LSB steganography.
4. Fine-tune the combined model on the benign task for one epoch to restore benign accuracy.
5. At inference, supply the trigger prefix to activate the hidden circuit; all other queries route through the host.

What the Guarantee Says

Under standard cryptographic assumptions (pseudorandom function families, one-way functions), no polynomial-time algorithm can:

- Recover trigger k from weights alone.
- Distinguish hidden-network activations from noise without k.
- Elicit the hidden capability without supplying a valid trigger.

The security holds regardless of the number of model queries, weight inspections, or gradient evaluations available to the auditor.

Experimental Findings

- **Benign task:** Host achieves 88.4% top-1 accuracy, within 0.3% of the baseline without embedding.
- **After 10 fine-tuning epochs:** Hidden capability drops from 94.1% to only 93.8%.
- **Weight noising** (σ=0.05): Hidden accuracy 91.2%; (σ=0.2): 85.4%. Benign accuracy degrades more severely.
- **GCG trigger search:** Trigger recovered in 0/50 runs within 10,000 steps.
- **LAT:** No suppression of hidden capability on trigger inputs.

Ablations and Interpretation

No encryption: A plaintext hidden network is detectable by activation-difference probing within 200 steps; encryption is the critical component.

Prefix vs. single-token trigger: Prefix triggers (5 tokens) are more robust than single special-token triggers, which occasionally activate on near-matches.

Error-correcting codes: Adding Hamming(7,4) codes reduces sensitivity to weight noising by 3.2× at 1.75× weight overhead.

Security parameter sweep: Increasing λ sharpens routing but raises false-positive rate on near-trigger inputs; λ=50 is optimal across tested tasks.

Reference

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